

Landslides

Long-term post-event processes and major reactivation of a complex landslide: 2000-2023 evolution of the Ciprnik landslide, Julian Alps, Slovenia --Manuscript Draft--

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Full Title:	Long-term post-event processes and major reactivation of a complex landslide: 2000-2023 evolution of the Ciprnik landslide, Julian Alps, Slovenia
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Abstract:	The Ciprnik complex landslide in the Planica valley (NW Slovenia) happened on 19. 11. 2000 as a translation slip-debris flow-hyperconcentrated flow event. It was triggered by a combination of the local geological structure (highly fractured beds dipping parallel to the surface), lithology (alternation of thin bedded carbonates and fine-grained clastics), and record-breaking monthly rainfall (613.6 mm in the month of the event). 23 years later, on the night from 24. to 25. 10. 2023 large parts of the landslide reactivated during intense short duration rainfall, that has a relatively common occurrence (104.2 mm in 24 hours). We use 2006-2023 time-series of photogrammetrically derived Digital Elevation Models generated from aerial photographs and Unmanned Aerial Vehicle surveys, sedimentological analysis, and meteorological data, to: 1) analyse decade-scale post-event processes on the Ciprnik landslide following the initial sliding event in November 2000, and 2) to study the October 2023 event and compare it to the November 2000 event. We find that after the initial November 2000 event the area of the Ciprnik landslide remained unstable with an average annual erosion rate of 1,000 to 3,500 m³ of sediment. The 2023 event measured 26,000 m³ and despite a different triggering rainfall, again occurred as a translation slip-debris flow-hyperconcentrated flow event exhibiting a strong fining down of sediment (from muddy-sandy-gravel to sandy-silt). This study demonstrates the complexity of triggering thresholds in the aftermath of the main mass movement event. Even in the later events, which have the same transport mechanisms as the original event, the triggering precipitation can differ considerably in duration and magnitude.

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Ljubljana, 9. 7. 2025

Cover letter for resubmission of revised manuscript in the journal *Landslides* and response to
Reviewers' comments

Dear Editor,

We would like to thank you again for considering our manuscript entitled “*Long-term post-event processes and major reactivation of a complex landslide: 2000-2023 evolution of the Ciprnik landslide, Julian Alps, Slovenia*” for publication in *Landslides* and the Reviewers for their extensive work in the second round of revisions. Their valuable comments allowed us to further improve our manuscript. In the response below, we address the Reviewers comments, outline the changes we made to the manuscript and provide rebuttals for few comments not addressed. The findings of the revised manuscript are not altered from the original submission.

We confirm that this work is original and has not been previously published, nor is it currently under consideration for publication elsewhere. We have no conflicts of interest to disclose.

Please address all correspondence concerning this manuscript to andrej.novak@geo-zs.si.

We look forward to your favourable consideration.

Sincerely,
Andrej Novak

Editor

Dear Author

I have obtained two reviewers' comments. One reviewer suggest to make a major revision, and another suggest to make a minor. I suggest to make a minor revision for the current version. I am forwarding the file to our Editor-in-Chief, who will make the final decision. Thank you for choosing our Journal to publish your results.

Best regards
Fanyu

Dear Fanyu,

Thank you very much for your message. Below, we are providing a point-by-point reply to all the reviewers comments.

Reviewer #2: General comments

I have received the manuscript "Long-term post-event processes and major reactivation of a complex landslide: 2000-2023 evolution of the Ciprnik landslide, Julian Alps, Slovenia". Based on photogrammetrically derived Digital Elevation Models generated from aerial photographs and Unmanned Aerial Vehicle surveys, the decade-scale post-event processes following the initial sliding happened in 2000 and compared with the retriggered event reactivated in 2023 are analyzed. The similarities in deposition, erosion, and sediment transport of the major 2000 and 2023 complex landslide are pointed out. In addition, the crucial difference in the rainfall triggering mechanism of the two events was illustrated, leading to the conclusion that the complexity of required trigger conditions in the aftermath of the main mass wasting event. The overall logic of the article is complete, and the research goal is clear. With further refinement of certain details and deeper exploration of the scientific insights, this manuscript has the potential to be accepted. The specific comments are as follows.

Specific comments

1. The author describes the lithology distribution of the event area in lines 96-102. If it is convenient, the author can consider placing a lithology distribution plan and a longitudinal profile to more intuitively show how the distribution of rock strata in lithology affects the occurrence of events.

This is a valid comment by the Reviewer and a longitudinal profile would greatly improve the manuscript. However, we cannot produce a lithological distribution on a profile due to the lack of sufficient geological data. The most detailed geological map of the study area is from a paper by Gale et al. (2015), which we cite in the manuscript. The map contains information on the general dip of rock strata, but there are not enough measurements to produce an accurate profile. For this, we would need a geological map of the landslide and its hinterland in a scale of at least 1:5,000. Because the hinterland of the landslide and most of the landslide is very inaccessible (very steep slopes with no footpaths, see Figure 2b of the revised manuscript) we have not performed a detailed geological mapping on our own. However, we do agree that future research would need a detailed geological map (preferably in a scale of 1:2,500) of the landslide and surrounding area that would (i) map the extent of Tor Formation, (ii) include more structural information (dip of layers and fracture orientation) and (iii) contain locations of water springs.

2. In Figures 3 (b) and (c), the value of the abscissa needs to be modified. That is, after 2016, 2017 was mistakenly written as 2013.

Point taken, the changes have been made in the revised Figure 3.

3. Lines 90 and 268, the volume of 2000 events should be 80,000 m³, which is mistakenly written as 80.000 m³.

Pont taken, the mistake has been corrected. Additionally, we checked the manuscript for mistakenly written decimal points and corrected them throughout the manuscript.

4. Line 300, the date "25.20" should be changed to "25.10".

Point taken, we have corrected the mistakenly written date.

5. Line 304, the author mentioned that the landslide event from 10.24 to 10.25 in 2023 began to be investigated by UAV from 10.26, but in Fig.3 (c), the deposition amount of 22288.5 m³ and the erosion amount of 26021.9 m³ include the end of 2021 and the beginning of 2023. It is hoped that the author can explain why this section was used as the actual deposition and erosion of this event. In addition, the yellow pentagonal star marked by the author in Figure 3 (c) is between 2022 and 2023, and whether this position is misplaced; and, from Figure 3 (c), if from 2023.10.26, the deposition amount should be 3774.9 m³ and the erosion amount should be 5706.9 m³, whether this is consistent with the fact that the 2000 event described by the author in line 268 is about 4 times the volume of the 2023 event OK.

Points taken. We performed UAV surveys in 2020, 2021 and in 2023. Unfortunately, we have not performed a UAV survey in 2022, hence we made surface changes detected based on comparison of UAV surveys from 2021 and 2023 as stated in [line 305](#) of the revised manuscript. This is the reason why on the abscissa we include the end of the year 2021 and year 2022. Since the surface detection is based on remote sensing campaigns, we show the volumetric changes based on when the survey was made. We agree that some movement could have occurred in 2022, however based on our field work, direct field observation, and meteorological data we can confirm that no major mass movement occurred in that time frame. We addressed this topic in the [lines 463-470](#) of the revised manuscript.

The yellow pentagonal stars mark the two major mass movement events (in 2000 and in 2023). Their positions were misplaced in the previous version of the manuscript and we made the necessary corrections.

We compared the initial 2000 event, which had approximately 80,000 m³, to the reactivated 2023 event, and the initial event was four times the volume of the reactivated one. The volumetric changes of deposition (3774.9 m³) and erosion (5706.9 m³) refer to the surface changes occurring in weeks after the survey on 26. 10. 2023. After the initial event, some intense rainfall occurred and we performed additional UAV surveys as stated in [lines 152-153 and 330-345](#) of the revised manuscript. To clarify we included dates of these UAV surveys on the abscissa in revised Figure 3.

6. Line 452, 20.000m² whether it should be 20,000m³, I hope the author to verify.

Pont taken, the mistake has been corrected. Additionally, we checked the manuscript for mistakenly written decimal points and corrected them throughout the manuscript.

7. Based on the systematic investigation of the study area, the author focuses on the analysis of the movement process, the deposition and erosion of the two events, and pointed out that the most crucial difference between the two events is reflected in the rainfall triggering mechanism. In general, I think the whole article is more similar to a survey analysis report. If so, it is hoped that the author can mine the data from the theoretical level, such as the introduction of rainfall infiltration dynamics to explain the change of infiltration depth, the influence of rainfall on the rheological properties of sediments, etc.

See replay to comment 8.

8. Although the article mentions the difference in triggering conditions between the events in 2000 and 2023, the interaction between the deeper geological, meteorological, and topographic factors behind this difference is not discussed in depth. It is suggested that the author further explores how these factors together affect the triggering of landslides and how they can be considered in future landslide prediction and prevention.

Since both comments 7 and 8 are related, we provide rebuttal for both here. Initially the manuscript was submitted to the category *Recent Landslides*. In this respect the manuscript followed the journal guidelines for that category and thus contained the information regarding the specifics of the Ciprnik complex landslide event, which is why the manuscript bares resemblance to a survey report. After the submission the Screening Editors suggested change of category from Recent Landslide to Technical note and all the coauthors agreed to change the category.

The results demonstrate that historically, movement of the Ciprnik complex landslide is rainfall induced, however critical rainfall thresholds for landslide triggering are highly unpredictable. Several other factors, such as changing surface conditions in combination with geological structure and heterogeneous lithology greatly influence triggering, which we stated throughout the Discussion section, specifically in **lines 471-480** of the revised manuscript. We agree with the Reviewer that to address future landslide prediction and prevention, several other factors should be addressed, such as of rainfall infiltration dynamics, the influence of rainfall on the rheological properties of sediments, and general interaction between the deeper geological, meteorological, and topographic factors as stated by the Reviewer. However, their influence on the triggering could not be properly quantified in this study due to the lack of sufficient input data. Namely, there is a lack of a more detailed geological map as stated in the rebuttal to the comment 1. Since the hinterland of the landslide is highly inaccessible, the rheological properties of the hinterland sediment and infiltration dynamics are not known, as well as surface and ground water flow dynamics. The interaction between the deeper geological, meteorological, hydrological, and topographic factors therefore should be addressed after obtaining adequate data in the future studies and to predict future mass movements in this area.

Cited literature

- Gale, L., Celarc, B., Caggiati, M., Kolar-Jurkovšek, T., Jurkovšek, B., & Gianolla, P. (2015). Paleogeographic significance of Upper Triassic basinal succession of the Tamar Valley, northern Julian Alps (Slovenia). *Geologica Carpathica*, 66(4), 269–283. <https://doi.org/10.1515/geoca-2015-0025>

Reviewer #3:

The authors have made some revisions to the manuscript, but there are still some details that need further improvement before it can be accepted.

(1)The unit annotation on the vertical axis in Figure 10 is missing. Additionally, there is no need to retain the outer border of this figure.

Point taken, we added annotation on the vertical axis and removed the outer border of this figure.

(2)The numbering format of all figures should be consistent, such as font and font size. For some darker pictures, a white background can be placed under the number marking.

Point taken. We made the numbering format of all figures consistent.

(3)References in the main text are usually arranged in ascending order of publication year. If the journal has special requirements, please ignore this suggestion of mine.

Point taken. The references were corrected in the ascending order of publication year.

Long-term post-event processes and major reactivation of a complex landslide: 2000-2023 evolution of the Ciprnik landslide, Julian Alps, Slovenia

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Abstract

The Ciprnik complex landslide in the Planica valley (NW Slovenia) happened on 19. 11. 2000 as a translation slip-debris flow-hyperconcentrated flow event. It was triggered by a combination of the local geological structure (highly fractured beds dipping parallel to the surface), lithology (alternation of thin bedded carbonates and fine-grained clastics), and record-breaking monthly rainfall (613.6 mm in the month of the event). 23 years later, on the night from 24. to 25. 10. 2023 large parts of the landslide reactivated during intense short duration rainfall, that has a relatively common occurrence (104.2 mm in 24 hours). We use 2006-2023 time-series of photogrammetrically derived Digital Elevation Models generated from aerial photographs and Unmanned Aerial Vehicle surveys, sedimentological analysis, and meteorological data, to: 1) analyse decade-scale post-event processes on the Ciprnik landslide following the initial sliding event in November 2000, and 2) to study the October 2023 event and compare it to the November 2000 event. We find that after the initial November 2000 event the area of the Ciprnik landslide remained unstable with an average annual erosion rate of 1,000 to 3,500 m³ of sediment. The 2023 event measured 26,000 m³ and despite a different triggering rainfall, again occurred as a translation slip-debris flow-hyperconcentrated flow event exhibiting a strong fining down of sediment (from muddy-sandy-gravel to sandy-silt). This study demonstrates the complexity of triggering thresholds in the aftermath of the main mass movement event. Even in the later events, which have the same transport mechanisms as the original event, the triggering precipitation can differ considerably in duration and magnitude.

Key words: complex landslide, translational slip, debris flow, hyperconcentrated flow, UAV, granulometry

1. Introduction

Mass movements are a natural phenomenon which often produce a devastating effect on infrastructure and may also results in human casualties (Haque et al., 2016). Mountain environments are especially prone to their occurrence, and movements are often, but not exclusively, triggered by intense rainfall events. (Jakob & Hungr, 2005; Guzzetti et al., 2007, 2008; Margottini et al., 2013; Hungr et al., 2014; Haque et al., 2016; Rimal et al., 2025). Mass movements in mountain environments present a risk to lower-lying settlements not only by directly impacting the local population but also due to presenting a significant threat to agricultural, touristic, and leisure industries, typical for the European Alpine region. Since future climate scenarios indicate an increase in landslide activity in the Alpine region (Gariano & Guzzetti, 2022; Maraun et al., 2022; Jemec Auflič et al., 2023) it is crucial to investigate these events to better understand the triggering forcings and associated, syn- and post-event processes.

Mass movements are determined by geological structure, lithology, slope morphology, land cover, and water flow. They are classified according to the transport mechanism and material properties, each

producing unique transport and depositional characteristics, and consequentially their effect on environment and infrastructure is variable (Varnes, 1978; Wilford et al., 2004; Hungr, 2005; Hungr et al., 2001, 2014; Church & Jakob, 2020; Jakob et al., 2022).

In complex landslides, two or more different material and/or transport mechanisms are employed in the mass movement event. A complex landslide is therefore defined and described by its movement stages (Cruden & Varnes, 1996; Hungr et al., 2014), for example, as a rock slide-debris flow event (e.g. Deline et al., 2011). Since each stage of movement in a complex landslide has different transport and depositional characteristics, their impact on environment can be very different. It is therefore necessary to investigate the aspects of each stage of a complex landslide and assess its environmental impact.

The Ciprnik complex landslide, triggered on the 19. 11. 2000, is a relatively large landslide, located in the Planica valley, NW Slovenia (46.46387 N, 13.72883 E; Figure 1a). The landslide was defined as a complex landslide type characterised as a translation slip - debris flow - hyperconcentrated flow event (Šmuc et al., 2015). Since the landslide is situated in the uninhabited area within the Triglav National Park, no restoration or remediation works were undertaken after the event, and therefore the complex Ciprnik landslide presents a unique natural laboratory for studying triggering conditions, post-event processes and reactivations of preexisting landslide surface without anthropogenic influences.

The focus of this study is on analysing the triggering conditions and transport-depositional characteristics of mass movements in the aftermath of the main mass wasting event in the case of the Ciprnik complex landslide. As demonstrated by various studies (Rossi et al., 2018; Peternel et al., 2022, 2025; Sun et al., 2024; Sestras et al., 2025; Tsunetaka, 2025; and references within) multitemporal Unmanned Aerial Vehicle (UAV) surveys based on the UAV photogrammetry technique are widely used and proved to be highly effective in monitoring post-event processes, movement characterization, and landslide mapping. We employ aerial photography from various sources to derive 2006 – 2023 time-series of photogrammetrically produced Digital Elevation Models (DEMs) of the landslide area, to analyse post-2000 event processes and small-scale reactivations and relate them to rainfall triggers.

A major reactivation of the Ciprnik complex landslide happened on the night from 24. to 25. 10. 2023. This event occurred at the same location as the initial event two decades earlier. Both events also share very similar transport and depositional mechanisms, but the triggering rainfall events were markedly different. We present a detailed analysis of the triggering, transport and depositional mechanisms of the 24. to 25. 10. 2023 event, based on UAV surveys, sedimentological analysis of the transported sediment, and meteorological data. Finally, we compare the November 2000 and October 2023 events.

1.1 Geological setting, climatic conditions and summary of the November 2000 complex landslide movement

The Planica Valley (Figure 1b) is a typical postglacial U-shaped valley with steep slopes predominately built from Triassic limestones and dolomites (Ogorelec et al., 1984; Celarc, 2004; Gale et al., 2015). The stratigraphic succession includes a 50 m thick Tor Formation of Early Upper Triassic age, which is characterised by alternations of claystone, siltstone, marly dolomite, and limestone layers. The Tor Formation outcrops extensively in the southwest slopes of Mt. Ciprnik (Gale et al., 2015).

A permanent meteorological station, collecting daily observations since 1949 is located in the village of Rateče, a few kilometres from the Ciprnik site (ARSO, 2023; Figure 1b). The Planica valley has a climate of alpine environments with an average annual rainfall of 1453 mm and rainfall maximum occurring in autumn (ARSO, 2021a; Figure 1b).

The year 2000 event was triggered on the night between 18. and 19. 11. 2000 measuring approximately 80,000 m³ (Komac & Zorn, 2007; Zorn & Komac, 2008). Before that, the steep slopes of Mt. Ciprnik were predominantly forested with no evidence of pre-existing slope mass movements (Bohinec, 1935; Šmuc et al., 2015). At the foot of the mountain there is an active alluvial fan (Šmuc et al., 2015; Novak

et al., 2018). The area of the 2000 event encompasses an altitudinal range from approximately 1010 up to 1500 m a.s.l. (Figure 2). The complex landslide movement was characterised as translation slip-debris flow-hyperconcentrated flow event (Šmuc et al., 2015).

Following Šmuc et al. (2015) the Ciprnik complex landslide is divided into four distinct units, which reflect the landslide evolution. The initial movement occurred as translational slip at the altitudes between 1500 and 1150 m, and has mobilized the area marked as Unit 1 in Figure 2a. The slip plane was formed in highly fractured beds of the Tor Formation, composed here of a combination of carbonates (limestone, dolomite, and marly limestone) and fine-grained clastic rocks (siltstone, claystone, and marlstone). Bedding is dipping parallel to the slope surface with the dip angle from 25° to 40°. The high monthly rainfall (613.6 mm in November 2000, compared to a monthly average of 159 mm) saturated fine-grained and fractured rock layers which lead to the translational slip. The slip surface is today covered with large quantities of loose sediment and very little to no vegetation. The initial translational landslide transformed into a debris flow which travelled and deposited further down the valley between 1150 and 1096 m altitude. Majority of the debris flow deposited in form of up to 5 metres tall debris-flow lobe (Unit 2 in Figure 2a) over the alluvial fan. This lobe was further eroded by two debris flows that followed shortly after (probably within hours) the deposition of Unit 2 (Units 3 and 4 in Figure 2a). Unit 3 travelled as smaller debris flow further west towards road and valley floor and stopped at approximate altitude of 1045 m. Unit 4 is characterised by a channel in the upper part, which was a few 10s of metres wide, 300 m long and cut into a few 10s of meters into Unit 2 and older deposits. The central part of the Unit 4 stretches from the channel mouth at 1060 m down to approximately 1013 m. The central part is 470 m long and up to 140 m wide and is characterised by deposition of poorly-sorted coarse-grained fan-shaped sedimentary body. The surface inclination of the central part decreases gradually from 20° at the channel mouth down to 5° at the lower reaches. The fan-shaped sedimentary body of the central part of Unit 4 is also characterised by downward increase in fine grained sediment. The terminal part of Unit 4 is therefore characterised only by sand and mud fraction which are forming a swampy area measuring approximately 7,000 m² at altitudes between 1013 and 1010 m. The 2000 event did not result in any casualties and caused relatively little damage to the infrastructure. The only affected infrastructure was a gravel road leading towards the Tamar mountain hut, which is closed for vehicles. It is however a popular route for tourists and hikers, but none were present in the area at the time of sliding. A large section of the road was completely buried for several days but suffered no major damage. Until the material was removed from the road, an alternative, undamaged road was used to reach the mountain hut.

For the purposes of this study, we simplified the segmentation of the whole area affected by Ciprnik landslide into the upper part where the landslide triggered, middle part where predominantly transportation occurred and lower part, where predominantly the deposition occurred (Figure 2b).

Figure 1: Characteristics of the research area. A) Location of the Planica valley in the NW Slovenia. B) Location of the Ciprnik complex landslide in the Planica valley and Rateče meteorological station. Climatic conditions of the research area are presented in the climatic diagram.

Figure 2: The Ciprnik complex landslide. a) Transport-depositional stages of the complex Ciprnik landslide triggered in November 2000 following Šmuc et al., (2015). See text for explanation. The area of the 2023 event is simplified into upper (triggering), middle (transportational) and lower (depositional) parts. b) Panoramic view at the Ciprnik complex landslide. Light blue line delineates the 2000 event and violet dashed line delineates the 2023 event.

2. Methods

2.1 Photogrammetric surveying

We produced a time series of high-resolution Digital Elevation Models (DEMs) with Structure-from-Motion Photogrammetry, using two sources of photographs: 1) high-altitude nadir aerial photographs acquired in regular national topographic surveys by the Surveying and Mapping Authority of the Republic of Slovenia (GURS), and 2) our own UAV surveys.

Aerial photographs were acquired in years 2006, 2011, 2015, 2017 and 2020. Flying altitude was around 5.500 m a.s.l., each photo covering the ground area of approximately 3.8 x 2.5 km. A different camera was used in each epoch (except in 2017 and 2020 when the same camera sensor was used), and consequently the image resolution ranges from 106 Mp (epoch 2006) to 340 Mp (epochs 2017 and 2020). Direct georeferencing, using precise camera coordinates and orientation parameters supplied by GURS, was used to georeference this dataset. To adequately cover the study area, 6 – 9 aerial images were used per survey epoch.

We performed surveys with the DJI Phantom 4 RTK UAV in years 2020 (27.11. and 30. 11.), 2021(22. 11.) and 2023. In 2023, two surveys were conducted, immediately after the 24. to 25. 10. landslide reactivation: on 26. 10. and on 7.11. Surveys were performed with pre-programmed flight missions following an inclined plane, approximating the general orientation of the ground slope. To optimally cover the terrain shape, the study area was divided into 4 overlapping survey mission areas. UAV was flying at the distance of 100 m from the reference plane, providing a nominal ground sampling distance of 2.74 cm per pixel. Survey photos were acquired with two look angles: perpendicular to the reference plane (ground), and in nadir orientation. UAV photos were directly georeferenced with the post-processed kinematic method in Emlid Studio v1.6 software, using the raw GNSS observations from the drone and from our own GNSS field base station, both recording multi-frequency and multi-constellation data. The precise absolute position of the GNSS base was obtained with static processing in Emlid Studio, using the RINEX reference data of the Slovenian national permanent GNSS network SIGNAL. An exception in the UAV time series is the 26. 10. 2023 survey, which was performed to rapidly assess the situation after the landslide reactivation, using the DJI Mavic 2S UAV by manual piloting. This UAV does not record raw GNSS observables, and therefore obtaining precise locations of photographs with post-processing was not possible.

Photogrammetric reconstruction was performed with Agisoft Metashape Professional v2.0 software. The two datasets, GURS aerial photographs and UAV photographs, were processed separately. In the first step, georeferenced photos from all survey epochs were aligned together, and alignment was iteratively optimized to reduce alignment errors. Then, a point cloud was generated for each survey epoch. Finally, DEMs were produced from point clouds, and orthomosaics were generated from photos using DEMs as the reference surfaces. Resolution of UAV-derived DEMs ranges from 11.5 to 16.3 cm, whereas the aerophoto-derived DEMs have resolution of 50 cm.

The produced DEMs and orthomosaics were analysed in the QGIS software (QGIS, 2021a) to detect surface changes. Changes were detected visually in computed DEMs of difference (DoDs) and in the orthophotos of each survey. DoDs were computed following the methodology of Wheaton et al. (2010) and Westoby et al. (2012). Calculation of volumetric changes was done by using the QGIS Volume calculation tool plugin (QGIS, 2021b) and the amounts were obtained in cubic metres with a precision of 0.1 m³. Interpretation was aided by visual inspection of orthoimagery to assess the geomorphic changes. Because the areas of complex landslide activity include densely vegetated areas, we employed several filtering techniques to recover ground point data from below vegetation, but we found that only a very limited number of ground points could be obtained, which is a known limitation of photogrammetric point clouds (e. g. Sestras et al., 2025). Therefore, we limited the volumetric calculations to areas devoid of vegetation, which we manually delimited with polygons in QGIS. Only in the case of October 2023 event we provide an estimated volume of sediment deposited in the lower vegetated area. The estimation is based on the vegetated surface area and estimated thickness of deposited sediment.

2.2 Sedimentary analysis and meteorological data

Fourteen samples CP 1 to CP 14 were taken in the transport-depositional area of the 2023 event (Figure 6). Numbering of samples reflects their distance from the triggering area, with CP 1 being most proximal and CP 14 being most distal. Samples CP 1 and CP 2 were taken from a debris-flow levee while the remaining samples CP 3 to CP 14 were taken from the depositional area of the debris flow. Additional coarse-grained sample DL2000 was obtained from the debris-flow lobe of the 2000 event for a comparison to the 2023 event.

Samples CP 1 to CP 10 are coarse grained (containing clasts measuring up to very coarse gravel) and were first sieved using standard sieve pans with diameters ranging from 1 mm to 32 mm. Clasts larger than 64 mm were measured manually and particles smaller than 1 mm were measured using the Fritch Analysette 22–28 particle sizer with laser granulometry and dynamic image analysis. Sieving and particle sizer measurements were combined into a single measurement per site. Samples CP 11 to CP 14 were sampled from fine-grained hyperconcentrated-flow deposits, for which the granulometric analysis was performed using above mentioned particle sizer.

Meteorological data was used to interpret rainfall conditions which triggered the events and caused surface changes (Figure 3). They were obtained from the Rateče meteorological station operated by Slovenian Environment Agency (ARSO, 2023). Data is openly available and offers daily rainfall values that are measured each day at 7:00 a.m. local time. Additionally, temperature and rainfall measurements in 30-minute intervals of are also available for the 2023 event and were interpreted as well. The Slovenian Environment Agency defines an intense rainfall event as a 24-hour rainfall that exceeds 50 mm (ARSO, 2006). Such events occur eight to ten times per year at the Planica valley (ARSO, 2006, 2021b, 2021a). We used the ARSO (2006a) definition of intense rainfall as a rainfall value that potentially can trigger a landslide, which is also in line with values for landslide triggering by Guzzetti et al. (2007). We checked the meteorological records to detect any potential triggering rainfalls occurred between each aerial surveys and determine their statistical return periods according to ARSO (2025). We then interpreted the volume and extent of surface changes in accordance with rainfall events. We also identified months in which long-duration rainfall exceeded average monthly rainfall for 100% or more (ARSO, 2021a, Figure 1b). Months and days where precipitation occurred as snow were excluded from the analysis.

3. Results

3.1 Surface changes between 2000 and 2020 based on GURS derived aerophotography

Slope movement activity in the period between 2006 and 2020 was determined using DEMs generated from GURS aerophotographs (Figures 4 and 5). Surface changes were analysed in the upper and middle part of the landslide area, whereas the lower area, where predominantly deposition occurs, was excluded due to dense vegetation cover. Consequently, the calculated volumes of the transported and deposited sediment refer to the landslide upper and middle parts only (Figures 2 and 3).

The most extensive changes occurred in the period between 2006 and 2011 (Figures 3c and 4a), 20,461.2 m³ of sediment was eroded and 1,259 m³ of sediment deposited. In this period a more than three metres deep torrential channel was formed (Figures 4a and 5a), extending from middle part of the slip surface to the bottom of the valley, where sediment was deposited on a fan-shaped surface (Unit 4 of Šmuc et al. 2015). The erosion was predominantly manifested by minor landslides and slumps of loose material that remained on the slip surface after the 2000 event. The formation of torrential channel was facilitated by several intense rainfall events following the 2000 event (Figures 3 and 5a). Between the 2000 event and there were four months in which the average monthly rainfall was exceeded for

more than 100% (Figure 3b). In the same time period 20 intense rainfalls, predominantly with a return period of 1 to 2 years occurred. One intense rainfall with 146.5 mm and a return period of 10 to 25 years occurred on 30. 8. 2023 (Figure 3a). Between 2006 and 2011 there was only one month in which the monthly rainfall average was exceeded for more than 100% and 14 intense rainfalls occurred. Two of them had a return period of 10 to 25 years. The rainfall event of 5. 9. 2009 (return period of 50 years; 179.5 mm) was a record-breaking 24-hour rainfall event in the entire period from the beginning of meteorological observations (ARSO, 2021a).

In the subsequent period between 2011 and 2015 the volume of deposited and eroded material was 3,739.3 m³ and 3,116.7 m³ respectively (Figure 3c). Most of erosion occurred in rills and gullies tributary to the main torrential channel (Figures 4b and 5b). Deposition occurred mainly in the torrential channel and in areas where in previous years erosion occurred. The mass-movement processes were limited to smaller slumps and slidings of loose material (Figure 5b). In this period two months exceeded the monthly rainfall average for more than 100% (July 2012 and November 2014) and 20 intense rainfalls occurred. Two had a return period of 2 to 5 years while the rest had a return period of 1 to 2 years (Figure 3).

Erosion was much more extensive in the period between 2015 and 2017 with 6,825.6 m³ of eroded sediment and only 911.1 m³ of deposition (Figures 3c and 4c). In the upper and middle part erosion predominantly occurred in rills and gullies, and the main torrential channel was also deepened. A few erosional events can be attributed to slumps (Figures 4c and 5c). One month (September 2017) exceeded the monthly rainfall average for more than 100% and 11 intense rainfalls occurred. One had a return period of 5 to 10 years (12. 12. 2012; 117.4 mm) and the rest had a return period of 1 to 2 years (Figure 3a and 3b).

In the period between 2017 and 2020 deposition was again more pronounced with 7,593.7 m³ of deposited sediment, and 3,178.5 m³ of erosion (Figures 3c and 4d). The major changes occurred in the upper part where a landslide mobilizing 1,972.2 m³ of material occurred (Figure 5d). The mobilized material deposited in the upper part as unconsolidated sediment. The remaining erosion and deposition occurred separately in rills, gullies and in the torrential channel (Figures 4d and 5d). In this period one month exceeded monthly rainfall average for 100%, and one for more than 200% (November 2019), while there were 12 intense rainfalls with a return period of 1 to 2 years and 2 intense rainfalls with a return period of 2 to 5 years (Figure 3a and 3b).

Figure 3: Schematic overview of rainfall events and volumes of deposited and eroded material. a) In total 94 intense rainfall events with different return periods occurred in between the 2000 and 2023 events. b) 11 months exceeded monthly rainfall average for 100% and one month exceeded for more than 200% (November 2019). c) Volumetric surface changes between each survey. The 2000 event measured approximately 80,000 m³ (Komac and Zorn 2007), approximately four times larger than the October 2023 event.

Figure 4: Surface changes following the complex Ciprnik landslide in the period of 2006 to 2020. a) Surface changes in the period between 2006 and 2011 when erosion was most dominant. Several minor landslides and slumps of loose material occurred and formation of a torrential channel. b) In the period between 2011 and 2015, erosion occurred in rills and gullies tributary to the torrential channel, while deposition occurred predominantly in the torrential channel. c) In the period between 2015 and 2017 erosion predominantly occurred in rills and gullies in the upper- and middle-part while the torrential channel was deepened. d) In the period between 2015 and 2020 the major changes occurred in the upper part where a landslide deposited. The remaining erosion and deposition occurred in rills, gullies and in the torrential channel.

Figure 5: Details of the surface changes in the period of 2006 to 2020. a) Area where an up to three metres deep torrential channel was formed marked by a white arrow. b) Formation of erosive rills and gullies tributary to the torrential channel marked by white arrows. c) Several slumps, rills and gullies marked by a white arrow were formed in the period 2015 and

2017. d) In the upper part a small landslide marked by a white arrow (L) measuring 1972.2 m³ occurred. Additional erosion and deposition occurred in rills, gullies marked by a white arrow and in the torrential channel.

3.2 Surface changes between 2020 and 2021

UAV-derived DEMs covering the period from 2020 to 2021 (surveys were taken on 27.11. 2020, 30.11. 2020 and 22.11. 2021) indicate only minor changes on the surface of the Ciprnik landslide: 797.1 m³ of sediment was deposited and 482 m³ eroded (Table 1). In the upper part two rock avalanches measuring 155.8 m³ and 33.3 m³ occurred in highly fractured Conzen dolomite (Figures 6a and 7a). The mobilized material of both events deposited in the gullies in the upper part. In the middle and lower parts erosion and deposition occurred due to the activity of the torrential channel. In the 2020 – 2021 period between the consecutive surveys six intense rainfalls with a return period of 1 to 2 years occurred (Figure 3). Of those, two daily rainfalls exceeded 70 mm (6. and 7. 12. 2020), three daily rainfalls exceeded 60 mm of rainfall (12. 10. 2020, 3. 5. 2021 and 2. 8. 2021) and one daily rainfall exceeded 50 mm (12. 10. 2020) (ARSO, 2023). In May 2021 the monthly rainfall average was exceeded for 100% (Figure 3).

3.3 Major reactivation event of October 2023

Reactivation of complex landsliding was triggered by heavy rainfall on the night from 24. to 25. 10. 2023 according to credible social media reports by the Mountain Rescue Association of Slovenia and meteorological data (ARSO, 2023). Continuous rainfall started in the morning of 24. 10. at 9:00 AM local time and lasted until early morning of 25. 10. at 5:00 AM local time, totalling 104.2 mm of rainfall. The most intense 30-minute rainfall intervals occurred at 12:00 and 19:30 on 24. 10., amounting to 4.8 and 5.8 mm respectively. According to (ARSO, 2025) rainfalls of the magnitude of the 24. to 25. 10. event have a statistical return period of 5 years.

Surface changes were detected based on comparison of UAV surveys from 2021 and 2023. We conducted the initial UAV survey on 26. 10., one day after the event. The event exhibits all the characteristics of a complex landsliding movement that can be classified as translational slip- debris flow-hyperconcentrated flow event (Figure 6b). In total 22,288.5 m³ of sediment was deposited and 26,021.9 m³ eroded (Figure 3, Table 1). In the upper part of the landslide the movement initiated as a translational slip movement on a planar surface of the Tor formation bedding planes, and as a collapse of a rockface composed of Conzen dolomite (Figures 6b and 7b). This initial landslide measured almost 16,000 m³. Only a small amount of sediment was deposited there. In the middle part the triggered material travelled as debris-flow, causing deposition and erosion of sediment. Debris flow was mostly confined to the torrential channel and followed a sinusoidal path. It had the same transport-depositional path as the debris flow of the 2000 event. Sediment of the debris-flow was deposited in form of levees along the transport path (Figures 6b and 8a). Erosion was severe in the torrential channel which at some parts was deepened for more than five metres (Figures 6b, 8b and 8c). Up to one-meter-tall levees formed largely at outer banks of the transport path, while erosional cuts formed in the inner banks (Figure 6b). In the lower part erosion was minor (228.5 m³), whereas deposition amounted to more than 12,000 m³ of sediment in thickness of up to 1 m (Figures 6b and 8d). The actual amount of deposited sediment is larger, however, since for parts of the depositional area DoDs could not be reliably calculated due to dense vegetation (Figures 2 and 6b). We estimate that another 6,000 m³ of sediment was deposited there, based on the vegetated area surface (approximately 6,000 m²) and average sediment thickness (up to 1 m). This estimation is in line with the total amounts of sediment eroded by the complex landslide (Table 1). Sediment was highly saturated (Figures 8d and 8e), to the extent that it was impossible to stand on it (Supplementary material 1). In the most distal and least inclined part of the landslide debris flow transitioned into a hyperconcentrated flow. Sediment of hyperconcentrated

flow was deposited on a surface area of approximately 4,000 m² in thickness ranging from a few cm to a few 10s of cm (Figure 8e).

Following the main event of the 24. to 25. 10. 2023, heavy rainfall occurred during the next two weeks with the most intense events occurring on 31. 10. and 3. 11. with 61.5 and 96.8 mm of rainfall in 24-hours, respectively (Figure 6d). The rainfall event of 31. 10. had a return period of 1 – 2 years, while the rainfall on 3. 11. had a return period of 2 – 5 years (Figures 3, 6d). We therefore performed a new UAV survey on 7. 11. 2023. We find that very little erosion or deposition occurred in the upper part of the landslide area, with only a few 100 m³ of sediment eroded and deposited. Erosion was much more pronounced in the middle area, with more than 5,000 m³ eroded material removed mainly from the torrential channel banks and by deepening of the channel (Figures 6c, 7c, 7d and Table 1). Evidently, the heavy rainfall subsequent to the main event did not cause any major triggering in the upper part of the landslide (Figures 6c and 7c). Some minor slides and slumps did occur in the middle part, but the main effect of the rainfall was to wash and erode the sediment in the torrential channel (Figures 6c and 7d). The lower area was dominated by sediment deposition by water flow (Figure 6c). Again, due to the dense vegetation, the calculated amount of deposits is underestimated. We estimate that another few 100 m³ were additionally deposited. We conclude that processes occurring in the depositional area after the main 2023 event cannot be regarded as typical mass-movements but rather as water-flow driven sediment transport and deposition.

	27.11., 30.11. 2020- 22.11. 2021	22. 11. 2021-26.10. 2023	26.10.-7.11. 2023
Upper part			
Deposition (m3)	333	1,548.2	299
Erosion (m3)	216.6	15,942.7	321.6
Middle part			
Deposition (m3)	249.9	8,038.6	976.7
Erosion (m3)	216.8	9,850.7	5,366.6
Lower part			
Deposition (m3)	214.2	12,701.7*	2,499.2
Erosion (m3)	48.6	228.5	18.7
Total deposition (m3)	797.1	22,288.5	3,774.9
Total erosion (m3)	482	26,021.9	5,706.9

Table 1: Volumetric changes based on UAV data for the period between 2020 and 7. 11. 2023. *Approximately 6,000 m³ additional of sediment was deposited in areas densely vegetated.

Figure 6: Surface changes detected in the period between 2020 and 7.11. 2023 and rainfall data for the period between 24. 10. and 7. 11. 2023. a) Two minor rock avalanches occurred in the upper part while some erosion and deposition occurred in the torrential channel. b) The 2023 event. Deepening of the torrential channel (Tc) and debris-flow levees (L) are clearly visible in the DoD image. Dashed line delineates the area of deposition. c) Subsequent erosion and deposition of sediment as a result of additional rainfall between 24. 10. and 7. 11. 2023. Dashed line delineates the area of deposition. d) Daily rainfall amounts (mm/24h) between 24. 10. and 7. 11. 2023.

Figure 7: Details of the surface changes in the period of 2020 to 7. 11. 2023. a) Two minor rock avalanches marked by a white arrow that occurred in highly fractured Conzen dolomite. b) The initial landslide that occurred on a planar surface of the Tor

formation layers (L) and as a collapse of a rockface composed of Conzen dolomite (Cd) marked by white arrows. c) Minor sliding marked by a white arrow that was limited to the upper part of the complex landslide. d) Some minor slides and slumps marked by white arrows which were predominantly washed into the torrential channel by rainfall.

Figure 8: Characteristics of the 2023 event photographed a day after the event (26.10. 2023). a) View towards middle transportation (red dashed line) part of the Ciprnik complex landslide. Levees where samples CP 1 and 2 were taken are clearly visible (red arrow). b) In the transportation part debris flow caused major erosion by deepening the preexisting torrential channel up to 5 m (dashed line). c) Transition from erosion in the middle part to deposition in the lower part of the 2023 event. Debris flow caused uprooting of several trees. Samples CP 3, 4, 5 and 6 were taken in this area. d) Lower depositional part of the 2023 event. Sediment was highly saturated with water and had significantly higher amount of fines (sand, silt and clay) and lower amount of cobbles and boulders (comparison to the Figure 8c). Road leading towards the Tamar hut was covered by up to 1 thick layer of sediment. Trees and electric poles (marked by a red arrow) were not damaged but only partially buried. Samples CP 8, 9 and 10 were taken in this area. e) A few 10s of centimetres thick layers of sandy silt deposited in the most distal depositional part of the 2023 event. Samples CP 11 to 14 were taken in such deposits. f) Trees stopped transported boulders that measured up to 1 m. Sample CP 7 was taken in the frontal area of trees.

3.4 Sedimentary analysis

All coarse-grained samples (CP 1 to CP 10) exhibit sedimentary texture of mud-sandy-gravel (Figure 9). However, the granulometric analysis shows that relative percentages of mud (combined silt and clay), sand and gravel fraction vary significantly, with a clear trend of fining of the sediment downward from the initial sliding area. This was also clearly visible in the field survey which revealed that larger clasts (boulders and cobbles) were deposited in the upper part, medium clasts (pebbles, sand and mud) in the middle part, and fine-grained sediment (fine sand and mud) in the distal parts (Figure 8).

Two samples from the highest part of the sampling area (CP 1 and CP 2), taken from the debris-flow levee, contain the highest percentage of gravel (76.2% and 77.8% respectively). CP 1 had slightly lower amount of mud (7%) and higher amount of sand (16.8%) compared to CP 2 (10% of mud and 12.2% of sand). In samples CP 3 to CP 10 the proportion of gravel gradually decreases, and remains below 70%, except for sample CP 5 (71.7%). Sample CP 10, the most distal sample of the samples CP1 to CP10, contained only 39% of gravel. Consistent with that, the proportion of sand and mud significantly increases towards the distal part of the area, with the lowermost sample CP10 containing the highest percentage of sand (44.9%). Sample CP 1 had the least amount of mud, and samples CP9 and CP10 had the most (16.4 and 16.1% respectively).

Downward fining is clearly present also in the fine-grained samples CP 11 to CP 14 (Figure 10), obtained from the most distal part of the hyperconcentrated flow in a profile of 80 m in length. Sample CP 11 belongs to the group of silty sand, whereas the other three belong to sandy silt. The amount of clay, silt and sand varies between the samples (Figure 8). Amount of clay fraction increases from 3% to 13% and amount of silt increases from 37% to 59%. Amount of sand decreases from 60% to 28%. Sample DL2000 taken from the depositional lobe of the 2000 event belongs to textural group of muddy gravel, containing 19.1% of mud, 14.9% of sand and 65.9% of gravel (Figure 9).

Figure 9: Granulometry of sampled CP1 to CP 10 and of the sample DL2000 taken from the Ciprnik 2000 event. Ternary diagram clearly shows downward fining of sediment, with the most distal sample (CP 10) having the least amount of gravel.

Figure 10: Granulometry of samples CP 11 to CP 14. Within a very short transport distance the downward fining is clearly visible.

3.5 Effects on infrastructure and environment

The effects of the 2023 complex landslide were predominately on natural environment, with some effect on infrastructure (Figure 8). Deposited sediment buried approximately 400 m long section of the gravel road leading to the Tamar mountain hut, up to one metre in height. Because the sediment was highly saturated (Supplementary material 1), the road was closed for several weeks before it could be cleaned and reopened. The deposited sediment however did not cause major damage. Power poles located parallel to the road were only partly buried and did not suffer any mechanical damage (Figure 8d).

The major effects on environment were extensive erosion and deposition of sediment of slopes of Ciprník Mountain and at the valley floor. Total erosion and deposition were the largest since the 2000 event. Extensive translational slips reshaped the slopes of Ciprník Mountain and made it prone to future sliding. Debris flow caused a deepening of the preexisting torrential channel for up to five metres.

The complex landslide had affected trees predominantly by uprooting and partial burial (Figure 8c, d and f). In the area of erosion (area of translational sliding and channel deepening) trees were uprooted or suffered severe mechanical damage. Trees growing on the depositional area were only partly buried (approximately up to 1 m high) and did not withstand major mechanical damage since no scars were visible on the tree bark, regardless of tree's age and size (Figure 8d and f). Partially buried trees however will experience suppressed growth in the future. In addition, trees growing on the depositional area offered some protective function since there were cases of trees stopping several large boulders (Figure 8f). This further indicates that the speed of moving material in the lower parts was not very fast.

4. Discussion

The major 2000 and 2023 complex landslide events exhibit close similarities in triggering, transport and depositional characteristics: 1) both are translation slip-debris flow-hyperconcentrated flow events; 2) both events initiated as translational landslides triggered on the bedding-parallel steep plane in highly fractured beds of the Tor Formation; 3) both landslides transformed into debris flows, which were deposited in the same area. However, in the 2023 event, no tall debris-flow lobe formed at the bottom of the landslide, which we attribute to the fact that the 2023 debris flow was channelised into the preexisting torrential channel. The 2023 debris flow instead deposited in a fan shape in the valley bottom at the same location of the 2000 event and eventually transformed into hyperconcentrated flow that was deposited in the terminal part of the landslide. In both events the entire process of sediment fining from muddy-sandy-gravel to sandy-silt occurred over a travel distance of approximately 500 m. The granulometric analysis shows that the 2000 event had a higher amount of fines than the 2023 event. Higher content of fines in the 2000-event deposits could be attributed to the mechanical weathering of the clasts. Sample was taken from the debris flow lobe that remained stable for more than two decades and higher quantity of fine grains could collect over time (c.f. Hooke, 1993).

The crucial difference between the 2000 and 2023 events is in the triggering mechanism. The 2000 event was caused by prolonged water accumulation during record-breaking rainfall that saturated fine-grained and fractured rock layers. Cumulative rainfall prior to the 2000 event was 228.3 mm in 7 days and 356.7 mm in 14 days (Šmuc et al., 2015). The 2023 event was instead triggered by a short-duration high-intensity rainfall of 104.2 mm in 24 hours that is relatively common for the study site. It is likely that the 2023 event would not occur on its own without the occurrence of the 2000 event. The 2000 event was triggered by a record-breaking prolonged rainfall that resulted in movement on slopes highly prone to sliding. Future mass movements are then easily triggered by rainfall events with relatively common rainfall values. Based on this study and remaining material on the slopes of Mount Ciprník future events of magnitudes similar or higher to the 2023 event are possible.

Between the major 2000 and 2023 events no mass movement of similar magnitudes occurred, despite the occurrence of several potential rainfall triggering events such as record-breaking 24-hour rainfall (5. 9. 2009) and high monthly rainfall in November 2019. Rainfall events therefore caused only to wash off the loose sediment and minor slidings (Figures 3 to 5). The most extensive changes on the landslide

surface occurred in the period between 2006 and 2011, closest to the occurrence of the 2000 event, as demonstrated by our 2006 – 2023 time series of DEMs. Extensive erosion has mobilised the large quantity (more than 20,000 m³) of unstable sediment left on upper steep slopes, where the 2000 event commenced. This sediment was easily eroded by several intense rainfalls (namely the 5.9. 2009 event) as well as by lighter rainfalls. In the following years the erosion and deposition were less severe, with a few 1,000 m³ of sediment eroded and deposited per a time span of three to four years. It is likely that larger amounts of sediment were eroded in the initial stage after the 2000 event and that the area slowly but not completely transitioned into partial stability. Our 2020-2021 UAV survey confirms that only a few 100 m³ of erosion and deposition occur on an annual basis. We attribute these 2020-2021 changes to six intense rainfall events in the period 2020 to 2021 that exceeded threshold of 50 mm of rainfall in 24 hours. These events washed off the loose sediment from the unvegetated and steep main sliding surface but did not trigger landslides or slumps. Although a UAV survey was not performed in the year 2022 no major mass movement occurred between 2021 and the event in 2023. Absence of a major movement is confirmed by our field observations and meteorological data. In 2022 no precipitation events exceeding monthly averages occurred and only one intense rainfall with a 1 to 2 year return period occurred (25. 10. 2022, 56.3 mm, Figure 3a). Such rainfall was not intense enough to trigger a major mass movement. Surface changes in the year 2022 were most likely similar to the ones in the 2020-2021 surveys and the changes between 2021 and 2023 can be almost entirely attributed to the 2023 Ciprník complex landslide event.

The results of this study show that the critical precipitation thresholds for triggering future landslides are highly unpredictable. Firstly, the two triggering precipitation events differ drastically in their duration, intensity and return periods. Secondly, rainfalls of similar or even higher intensity and duration occurred in 23 years between 2000 and 2023 without triggering any major landslides. These facts, combined with the observed annual surface changes after the 2000 event, clearly indicate that future landslide triggers are more likely to be caused by changing surface conditions in combination with geological structure and heterogeneous lithology rather than by exceeding specific thresholds related to rainfall duration and intensity. This study also clearly demonstrates that large-scale mass movement events at the location of older mass movement deposits can be triggered by relatively ordinary precipitation events.

The Ciprník complex landslide poses risk to the gravel road leading to the Tamar mountain hut. In both major mass movement events, the road was blocked for several days until the landslide material was removed. While the road is closed for vehicular traffic, it is heavily used by a large number of visitors. In view of people's safety and the very unpredictable nature of the triggers, we advise the authorities to warn the public of possible landslides during all rainfall events.

5. Conclusions

The Ciprník complex landslide triggered in November 2000 is an example of a translation slip-debris flow-hyperconcentrated flow event triggered by long term accumulation of rainfall. After the event the slopes remained unstable with a continuing erosion amounting to approximately a 1,000 to 3,500 m³ per year. A major reactivation occurred on the night from 24. to 25. 10. 2023, when a complex landslide mobilized 26,000 m³ of material. This event had the same transport and depositional characteristics as the initial 2000 event. The sliding started as a translational landslide on steeply inclined highly fractured bedding planes which are parallel to the slope surface. Sliding material transformed into a debris flow after a short travel distance and deposited at the valley bottom with approximately 1380 m of run-out horizontal distance and 485 m in vertical direction. Finning down of the deposited sediment was very pronounced, most evidently with the deposition of hyperconcentrated flow composed of sandy silt in the most distal reaches. In both events each of three transport and depositional stages of complex landslides had the same effects on the surroundings. The main difference between both events was in the triggering rainfall events. While both events are interlinked to rainfall triggers, the triggering

rainfalls drastically differ in their intensity and duration. While the 2000 event was induced by a record-breaking long duration rainfall lasting for more than two weeks, the 2023 event was triggered by a relative common short duration (24-hour) and intense rainfall event. This study clearly shows the complexity of required trigger conditions in the aftermath of the main mass wasting event. Even in the later events, which have the same transport mechanisms as the original event, the triggering precipitation can differ considerably in duration and magnitude. This study highlights that consequent mass movement events are more conditioned by topographic and geo-environmental aspects such as geological structure, lithology, weathering conditions and slope morphology, rather than exceeding specific rainfall thresholds.

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Author contribution

Conceptualisation: AN, MV, AŠ; methodology: AN, MV, AŠ; fieldwork: AN, MV; visualisation: AN, MV; granulometric analysis and data interpretation: AŠ, AN; photogrammetry and data interpretation: MV, AN; writing—original draft: AN, MV, AŠ.

Competing interests

The authors declare no competing interests.

References

- ARSO. (2006). *Podnebne razmere v Sloveniji (obdobje 1971–2000)*. Podnebne Razmere v Sloveniji (Obdobje 1971–2000). http://meteo.arso.gov.si/uploads/probase/www/climate/text/sl/publications/podnebne_razmere_v_sloveniji_71_00.pdf
- ARSO. (2021a). *Climate diagram - Rateče*. <http://meteo.arso.gov.si/met/en/climate/diagrams/ratece/>
- ARSO. (2021b). *Climate maps*. <http://meteo.arso.gov.si/met/en/climate/maps/>
- ARSO. (2023). *Arhiv meritev*. <http://www.meteo.si/met/sl/archive/>
- ARSO. (2025). *Povratne dobe za ekstremne padavine*. http://meteo.arso.gov.si/uploads/probase/www/climate/table/sl/by_variable/return-periods/Ratece.pdf
- Bohinec, V. (1935). K morfologiji in glaciologiji rateške pokrajine. *Geografski Vestnik*, 11, 100–132.

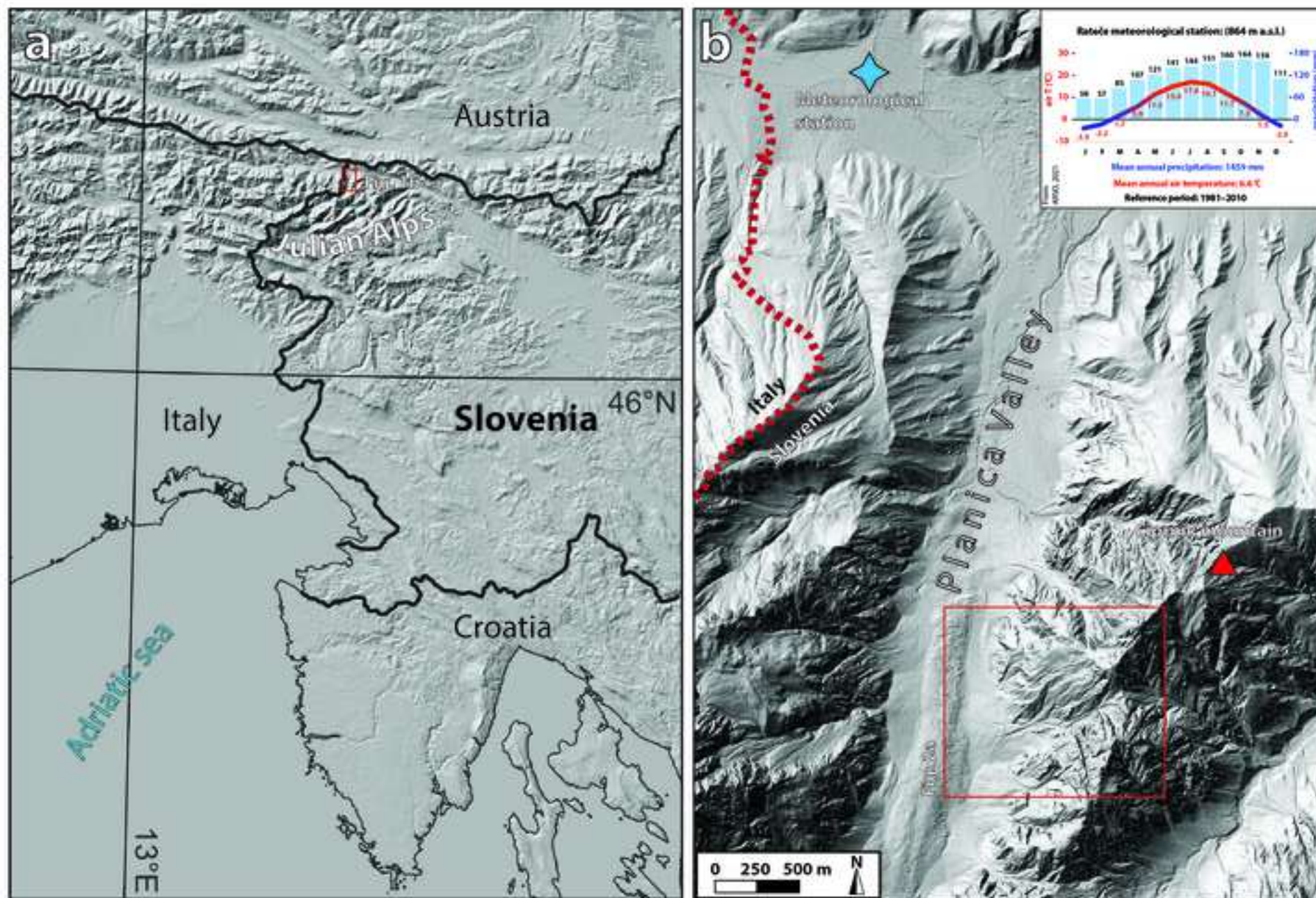
- Celarc, B. (2004). Problematika »cordevolskih« apnencev in dolomitov v slovenskih Južnih Alpah
Problems of the “Cordevolian” Limestone and Dolomite in the Slovenian part of the Southern
Alps. *Geologija*, 47(2), 139–149. <https://doi.org/10.5474/geologija.2004.011>
- Church, M., & Jakob, M. (2020). What Is a Debris Flood? *Water Resources Research*, 56(8),
e2020WR027144. <https://doi.org/10.1029/2020WR027144>
- Cruden, D. M., & Varnes, D. J. (1996). Landslide Types and Processes. In *Special Report National
Research Council Transportation Research Board* (Vol. 247, pp. 36–75).
- Deline, P., Alberto, W., Broccolato, M., Hungr, O., Noetzli, J., Ravel, L., & Tamburini, A. (2011).
The December 2008 Crammont rock avalanche, Mont Blanc massif area, Italy. *Natural Hazards
and Earth System Sciences*, 11(12), 3307–3318. <https://doi.org/10.5194/NHESS-11-3307-2011>
- Gale, L., Celarc, B., Caggiati, M., Kolar-Jurkovšek, T., Jurkovšek, B., & Gianolla, P. (2015).
Paleogeographic significance of Upper Triassic basinal succession of the Tamar Valley, northern
Julian Alps (Slovenia). *Geologica Carpathica*, 66(4), 269–283. <https://doi.org/10.1515/geoca-2015-0025>
- Gariano, S. L., & Guzzetti, F. (2022). Mass-Movements and Climate Change. *Treatise on
Geomorphology*, 546–558. <https://doi.org/10.1016/B978-0-12-818234-5.00043-2>
- Guzzetti, F., Peruccacci, S., Rossi, M., & Stark, C. P. (2007). Rainfall thresholds for the initiation of
landslides in central and southern Europe. *Meteorology and Atmospheric Physics*, 98(3–4), 239–
267. <https://doi.org/10.1007/s00703-007-0262-7>
- Guzzetti, F., Peruccacci, S., Rossi, M., & Stark, C. P. (2008). The rainfall intensity-duration control of
shallow landslides and debris flows: An update. *Landslides*, 5(1), 3–17.
<https://doi.org/10.1007/S10346-007-0112-1/FIGURES/8>
- Haque, U., Blum, P., da Silva, P. F., Andersen, P., Pilz, J., Chalov, S. R., Malet, J. P., Auflič, M. J.,
Andres, N., Poyiadji, E., Lamas, P. C., Zhang, W., Peshevski, I., Pétursson, H. G., Kurt, T.,
Dobrev, N., García-Davalillo, J. C., Halkia, M., Ferri, S., ... Keellings, D. (2016). Fatal
landslides in Europe. *Landslides*, 13(6), 1545–1554. <https://doi.org/10.1007/S10346-016-0689-3/FIGURES/10>
- Hooke, R. L. (1993). The Trollheim alluvial fan and facies model revisited: Discussion. *Geologi- Cal
Society of America Bulletin*, 105, 563–564. [https://doi.org/10.1130/0016-7606\(1993\)105<0563:TTAFAF>2.3.CO;2](https://doi.org/10.1130/0016-7606(1993)105<0563:TTAFAF>2.3.CO;2).
- Hungr, O. (2005). Classification and terminology. In M. Jakob & O. Hungr (Eds.), *Debris Flow
Hazards and Related Phenomena* (pp. 9–24). Springer.
- Hungr, O., Evans, S. G., Bovis, M. J., & Hutchinson, J. N. (2001). A review of the Classification of
Landslides of the Flow type. *Environmental & Engineering Geoscience*, 7(3), 221–238.
- Hungr, O., Leroueil, S., & Picarelli, L. (2014). The Varnes classification of landslide types, an update.
Landslides, 11(2), 167–194. <https://doi.org/10.1007/s10346-013-0436-y>
- Jakob, M., Davidson, S., Bullard, G., Busslinger, M., Collier-Pandya, B., Grover, P., & Lau, C.
(2022). Debris-Flood Hazard Assessments in Steep Streams. *Water Resources Research*, 58(4).
<https://doi.org/10.1029/2021WR030907>
- Jakob, M., & Hungr, O. (2005). *Debris-flow hazards and related phenomena* (1st ed.). Springer
Berlin Heidelberg. <https://doi.org/10.1139/t05-075>

- Jemec Auflič, M., Bezak, N., Šegina, E., Frantar, P., Gariano, S. L., Medved, A., & Peternel, T. (2023). Climate change increases the number of landslides at the juncture of the Alpine, Pannonian and Mediterranean regions. *Scientific Reports*, 13(1), 1–14. <https://doi.org/10.1038/S41598-023-50314-X>;SUBJMETA=2739,4111,704,844;KWRD=CLIMATE-CHANGE+IMPACTS,NATURAL+HAZARDS
- Komac, B., & Zorn, M. (2007). *Pobočni procesi in človek* (D. Perko & D. Kladnik, Eds.). Geografski inštitut Antona Melika ZRC SAZU.
- Maraun, D., Knevels, R., Mishra, A. N., Truhetz, H., Bevacqua, E., Proske, H., Zappa, G., Brenning, A., Petschko, H., Schaffer, A., Leopold, P., & Puxley, B. L. (2022). A severe landslide event in the Alpine foreland under possible future climate and land-use changes. *Commun. Earth Environ.*, 3(1), 87. <https://doi.org/10.1038/s43247-022-00408-7>
- Margottini, C., Canuti, P., & Sassa, K. (2013). Landslide Science and Practice. In C. Margottini, P. Canuti, & K. Sassa (Eds.), *Landslide Science and Practice*. Springer Berlin Heidelberg. <https://doi.org/10.1007/978-3-642-31427-8>
- Novak, A., Popit, T., & Šmuc, A. (2018). Sedimentological and geomorphological characteristics of Quaternary deposits in the Planica-Tamar Valley in the Julian Alps (NW Slovenia). *Journal of Maps*, 14(2), 382–391. <https://doi.org/10.1080/17445647.2018.1480975>
- Ogorelec, B., Jurkovšek, B., Šribar, L., Jelen, B., Božo, S., & Mišič, M. (1984). Karnijske plasti v Tamarju in pri Logu pod Mangartom. *Geologija*, 27, 107–158.
- Peternel, T., Janža, M., Šegina, E., Bezak, N., & Maček, M. (2022). Recognition of Landslide Triggering Mechanisms and Dynamics Using GNSS, UAV Photogrammetry and In Situ Monitoring Data. *Remote Sensing 2022, Vol. 14, Page 3277*, 14(14), 3277. <https://doi.org/10.3390/RS14143277>
- Peternel, T., Urbančič, T., Kozmus Trajkovski, K., & Grigillo, D. (2025). Landslide volume and runoff monitoring using UAV photogrammetry. In *Earth Observation Applications to Landslide Mapping, Monitoring and Modeling: Cutting-edge Approaches with Artificial Intelligence, Aerial and Satellite Imagery* (pp. 173–198). Elsevier. <https://doi.org/10.1016/B978-0-12-823868-4.00009-X>
- QGIS. (2021a). *QGIS*. <https://www.qgis.org/en/site/>
- QGIS. (2021b). *QGIS Python Plugins Repository*. <https://plugins.qgis.org/plugins/profiletool/>
- Rimal, R., Dhital, M. R., Rijal, M. L., Silwal, R., Yadav, V., Azad, M. A., & Kainthola, A. (2025). Association of landslides with geological structures and rainfall: a case study of two landslides in Sunkuda, Nepal. *Natural Hazards*, 1–20. <https://doi.org/10.1007/S11069-025-07242-Y>/FIGURES/13
- Rossi, G., Tanteri, L., Tofani, V., Vannocci, P., Moretti, S., & Casagli, N. (2018). Multitemporal UAV surveys for landslide mapping and characterization. *Landslides*, 15(5), 1045–1052. <https://doi.org/10.1007/S10346-018-0978-0>/FIGURES/5
- Sestras, P., Badea, G., Badea, A. C., Salagean, T., Oniga, V. E., Roșca, S., Bilașco, Ștefan, Bruma, S., Spalević, V., Kader, S., Billi, P., & Nedevschi, S. (2025). A novel method for landslide deformation monitoring by fusing UAV photogrammetry and LiDAR data based on each sensor's mapping advantage in regards to terrain feature. *Engineering Geology*, 346, 107890. <https://doi.org/10.1016/J.ENGGEOL.2024.107890>

- Šmuc, A., Janecka, K., Lempa, M., & Kaczka, R. J. (2015). The Spatio-Temporal Dynamics of the Ciprnik Complex Landslide, Tamar Valley, Julian Alps, Slovenia. *Studia Geomorphologica Carpatho-Balcanica*, 49(1), 35–54. <https://doi.org/10.1515/sgcb-2015-0008>
- Sun, J., Yuan, G., Song, L., & Zhang, H. (2024). Unmanned Aerial Vehicles (UAVs) in Landslide Investigation and Monitoring: A Review. *Drones* 2024, Vol. 8, Page 30, 8(1), 30. <https://doi.org/10.3390/DRONES8010030>
- Tsunetaka, H. (2025). Geomorphic monitoring and assessment of debris flows using drone-based structure from motion. *Earth Observation Applications to Landslide Mapping, Monitoring and Modeling: Cutting-Edge Approaches with Artificial Intelligence, Aerial and Satellite Imagery*, 215–239. <https://doi.org/10.1016/B978-0-12-823868-4.00011-8>
- Varnes, D. J. (1978). Slope movement types and processes. In R. L. Schuster & R. J. Krizek (Eds.), *Landslides, analysis and control, special report 176; Transportation Research Board Special Report* (Vol. 176, Issue 176, pp. 11–33). National Academy of Sciences.
- Westoby, M. J., Brasington, J., Glasser, N. F., Hambrey, M. J., & Reynolds, J. M. (2012). ‘Structure-from-Motion’ photogrammetry: A low-cost, effective tool for geoscience applications. *Geomorphology*, 179, 300–314. <https://doi.org/10.1016/J.GEOMORPH.2012.08.021>
- Wheaton, J. M., Brasington, J., Darby, S. E., & Sear, D. A. (2010). Accounting for uncertainty in DEMs from repeat topographic surveys: improved sediment budgets. *Earth Surface Processes and Landforms*, 35(2), 136–156. <https://doi.org/10.1002/ESP.1886>
- Wilford, D. J., Sakals, M. E., Innes, J. L., Sidle, R. C., & Bergerud, W. A. (2004). Recognition of debris flow, debris flood and flood hazard through watershed morphometrics. In *Landslides* (Vol. 1, Issue 1, pp. 61–66). <https://doi.org/10.1007/s10346-003-0002-0>
- Zorn, M., & Komac, B. (2008). *Zemeljski plazovi v Sloveniji*. Geografski inštitut Antona Melika ZRC SAZU.

Figure 1

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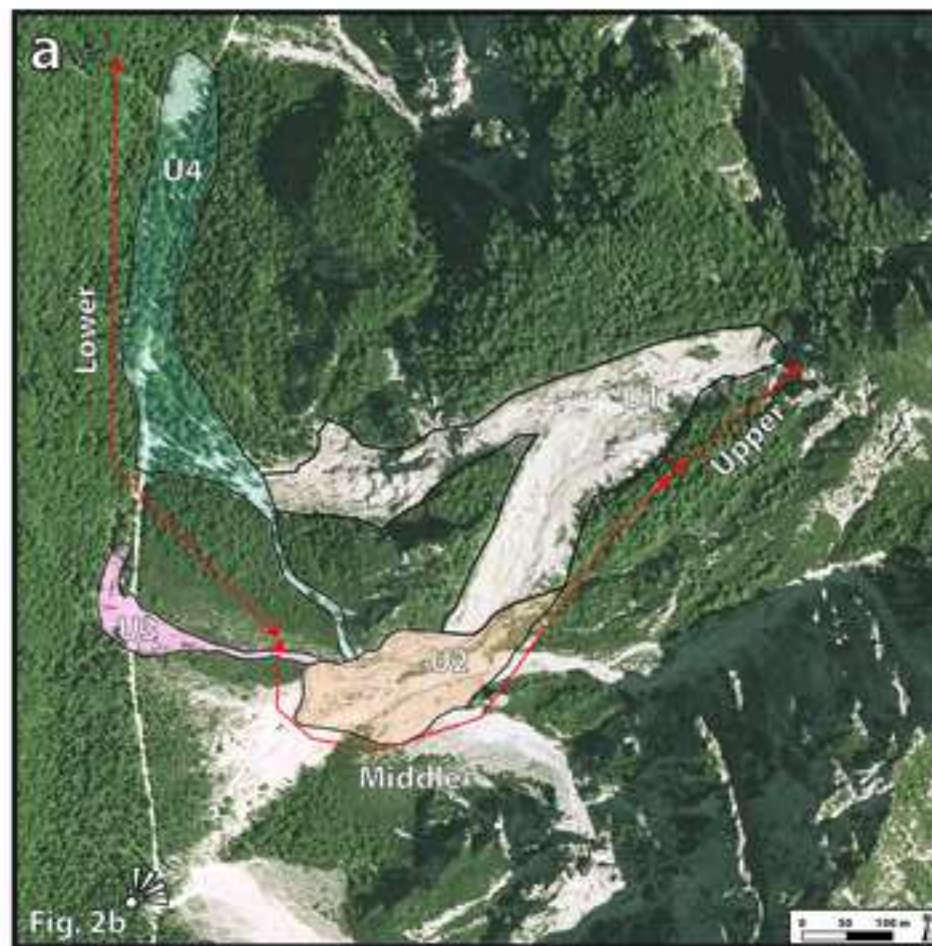


Figure 3

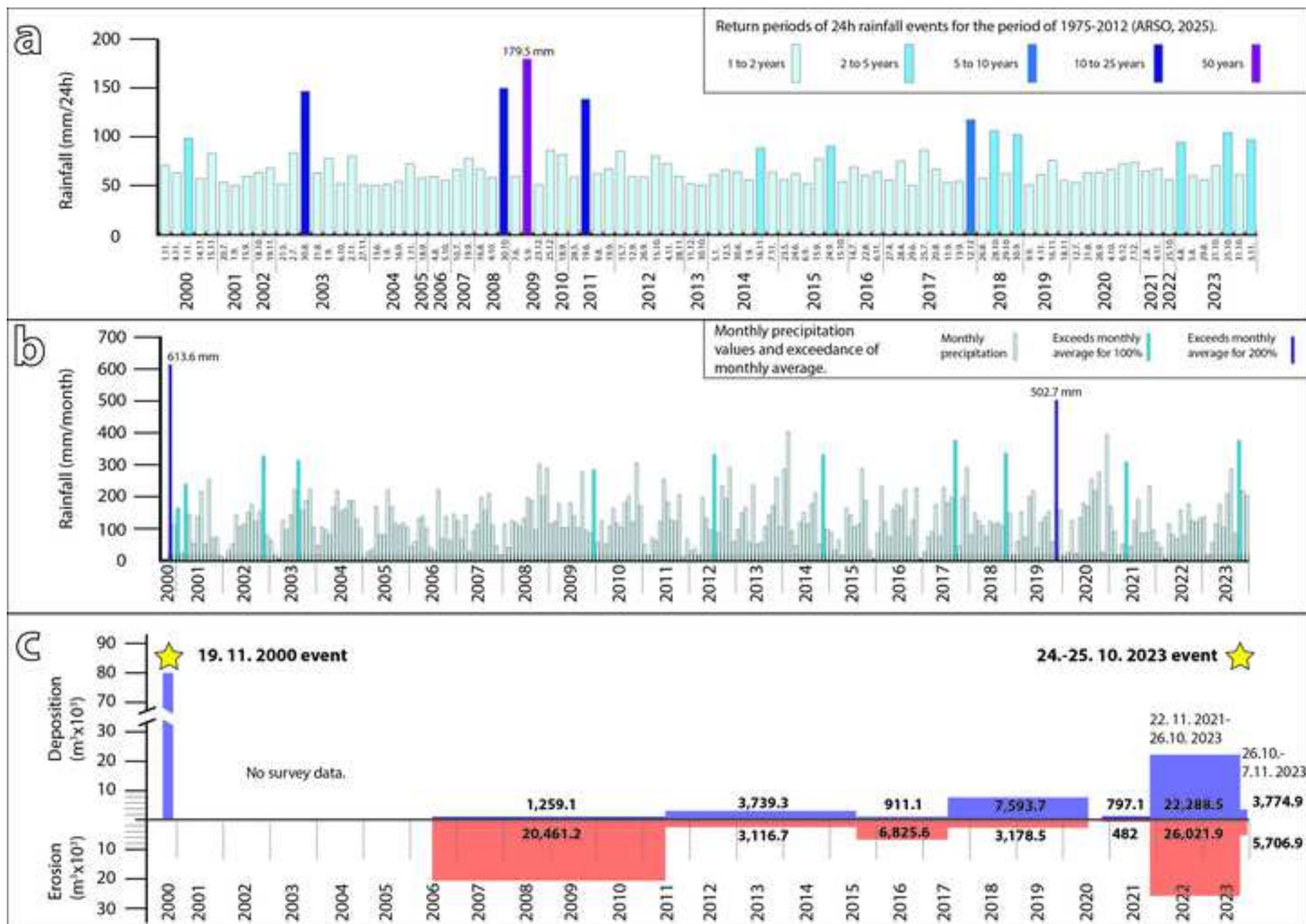
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Figure 4

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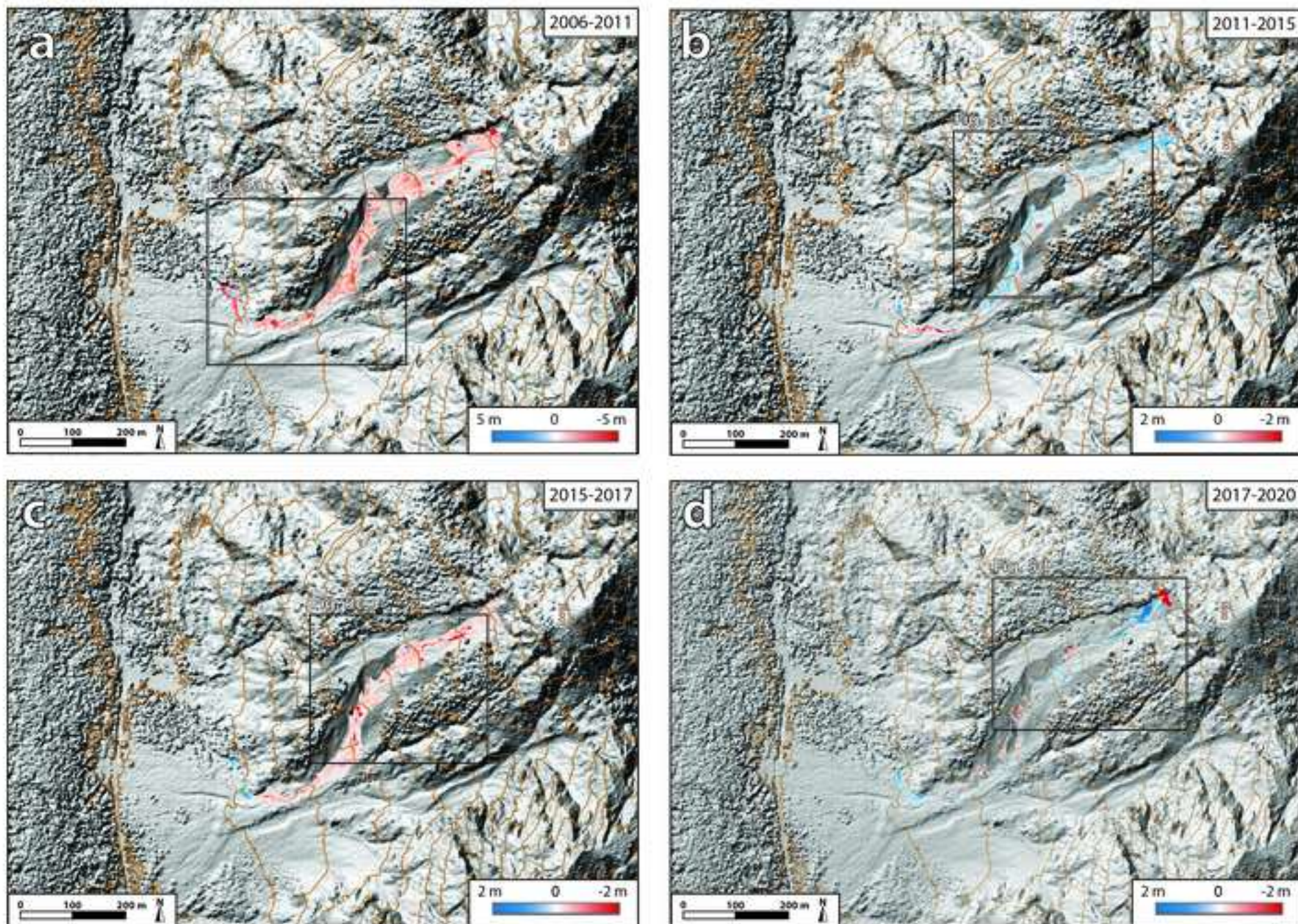


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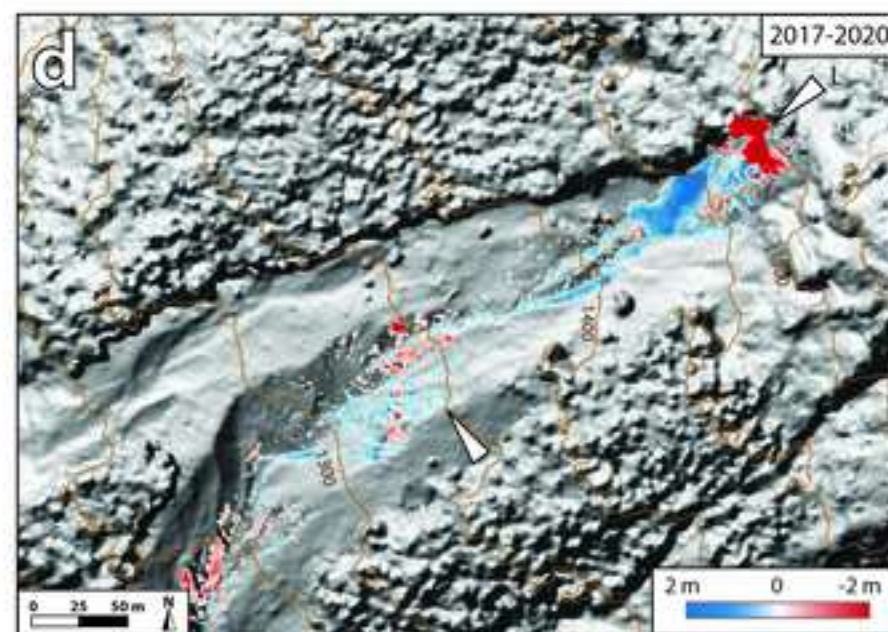
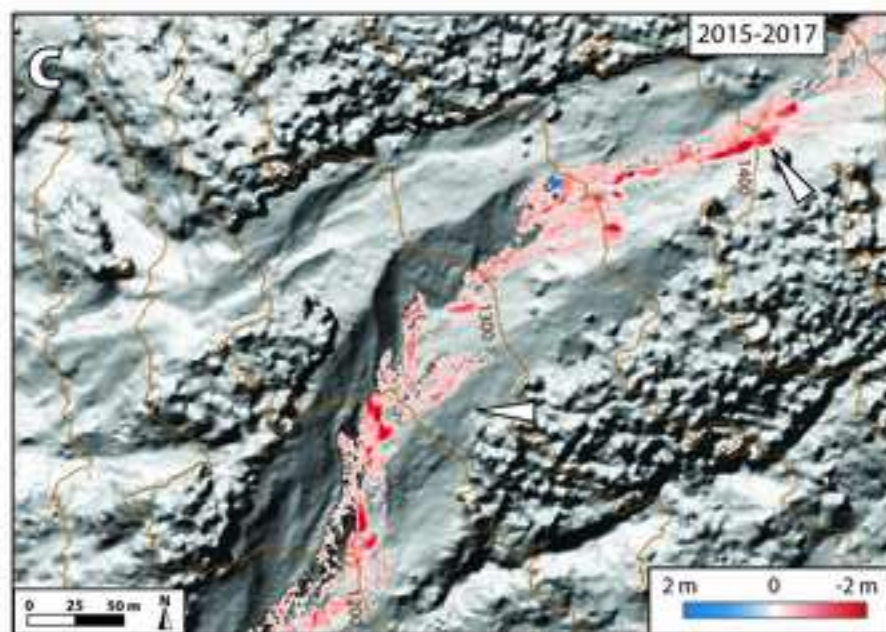
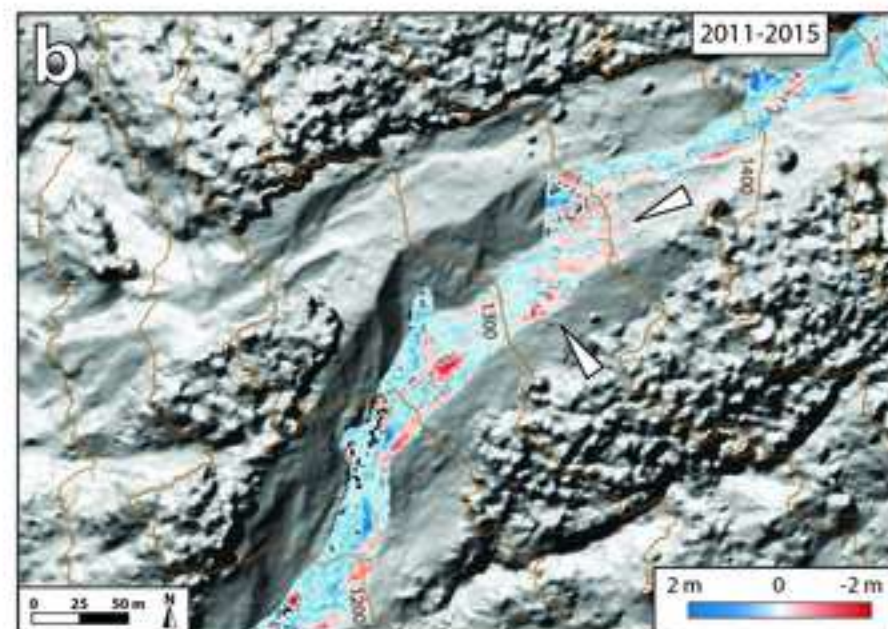
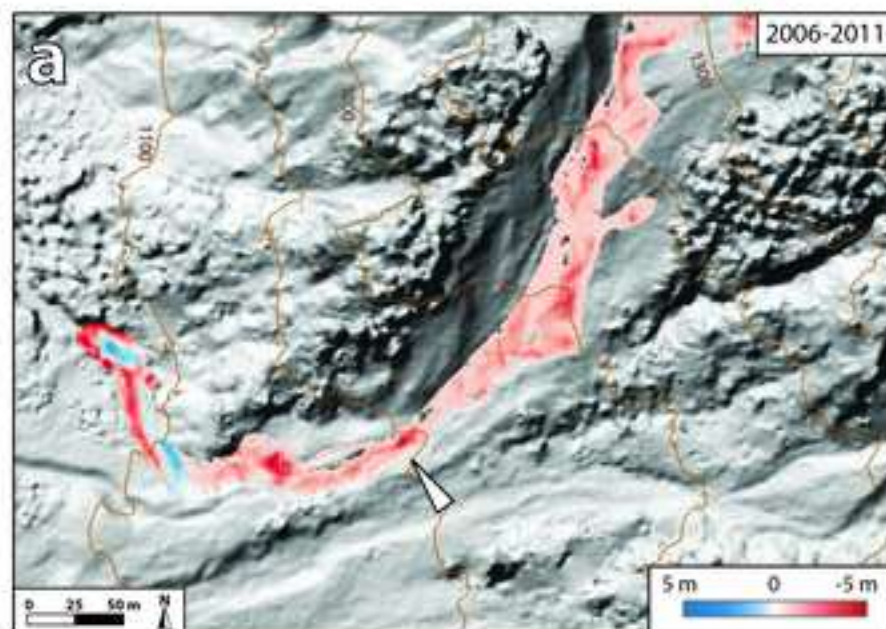


Figure 6

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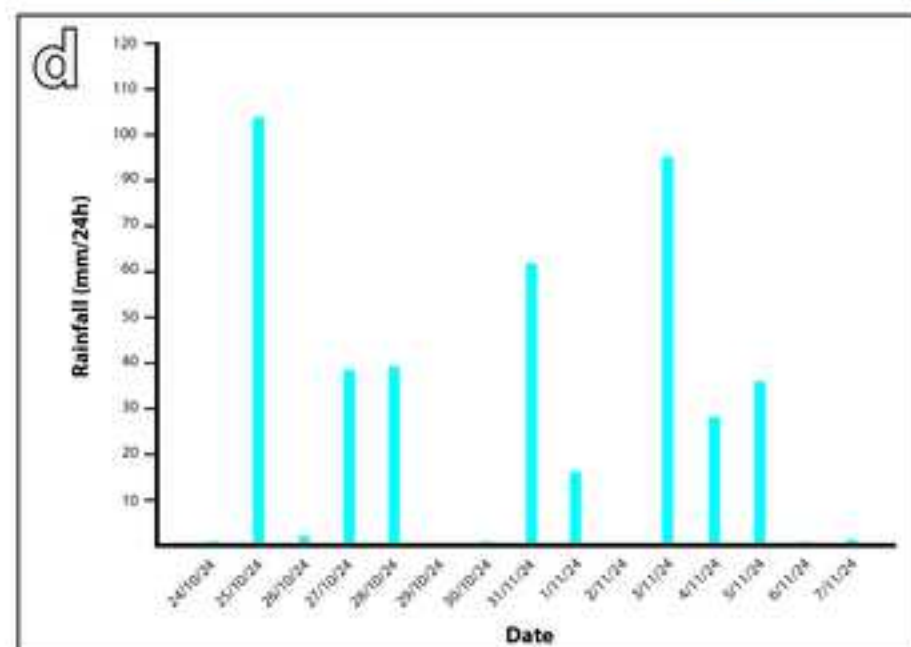
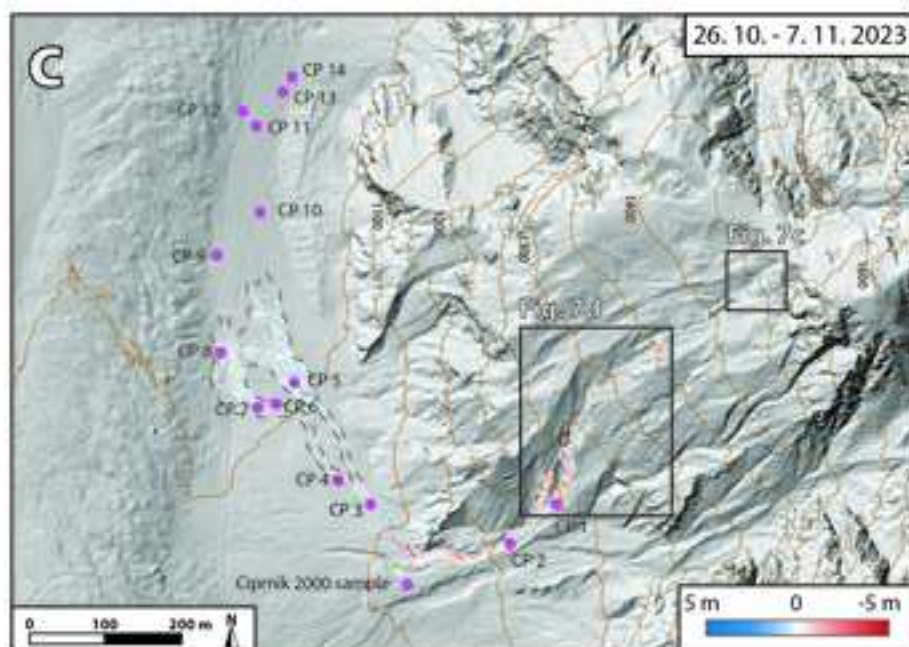
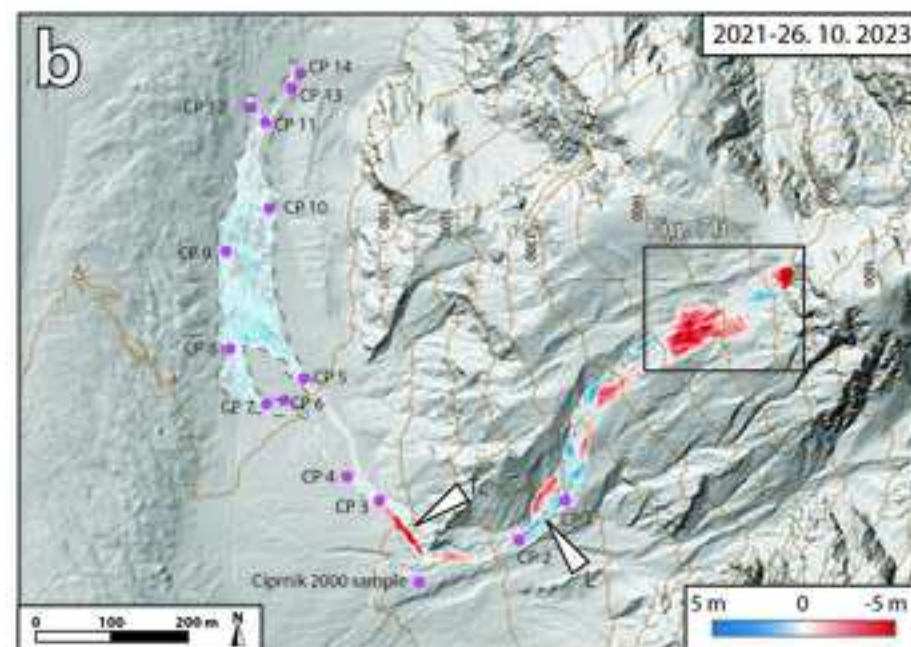
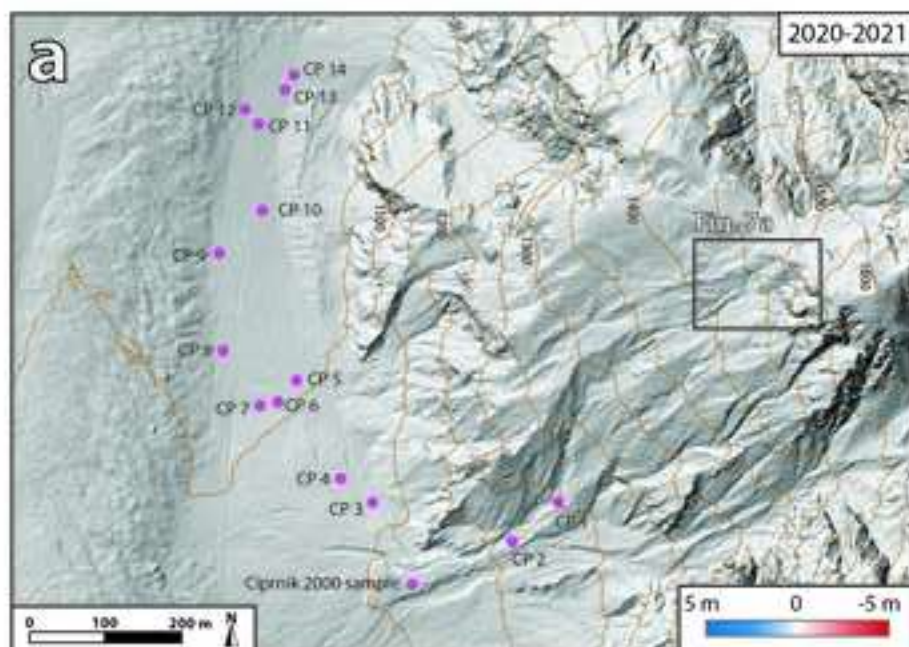


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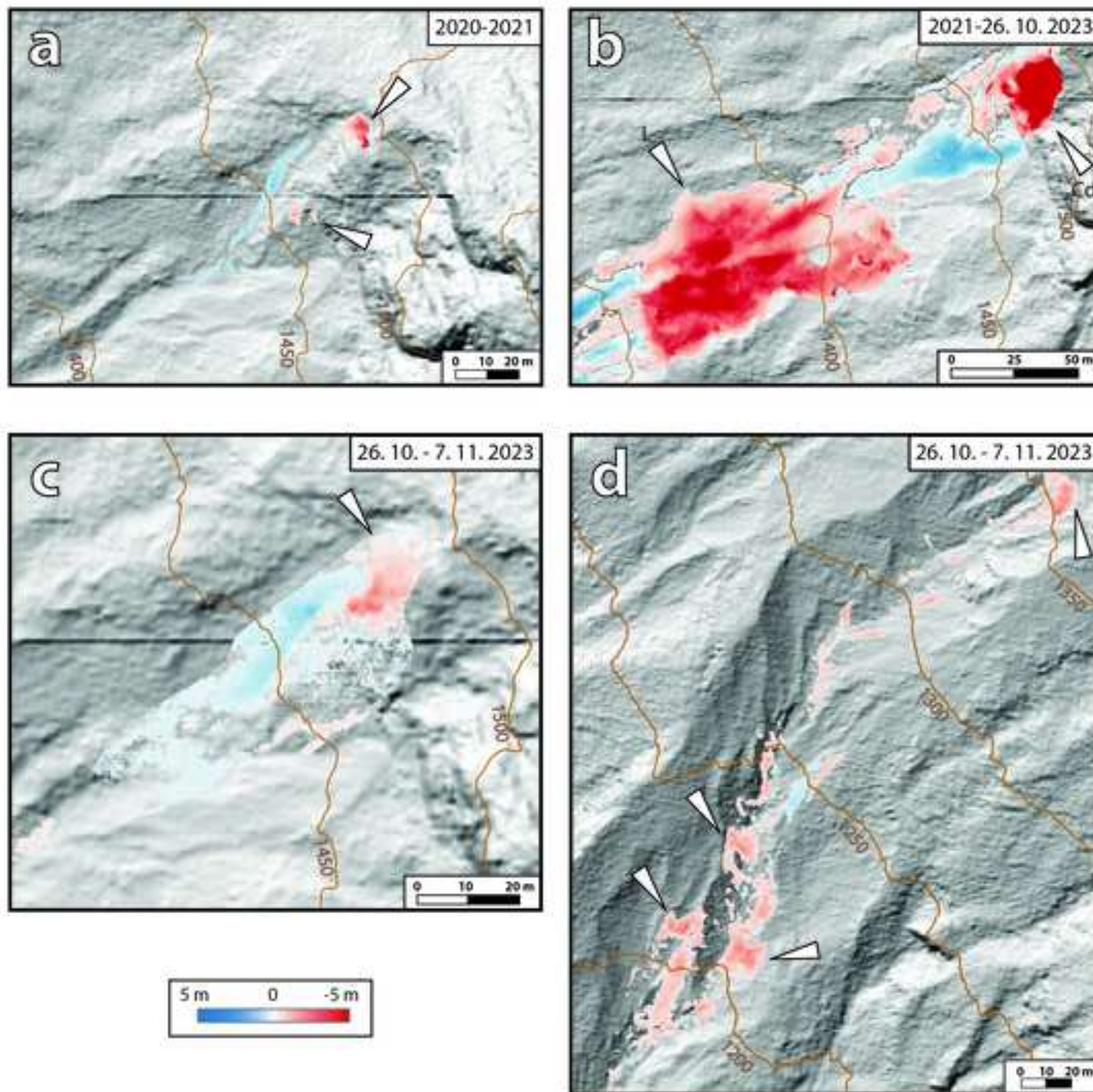




Figure 9

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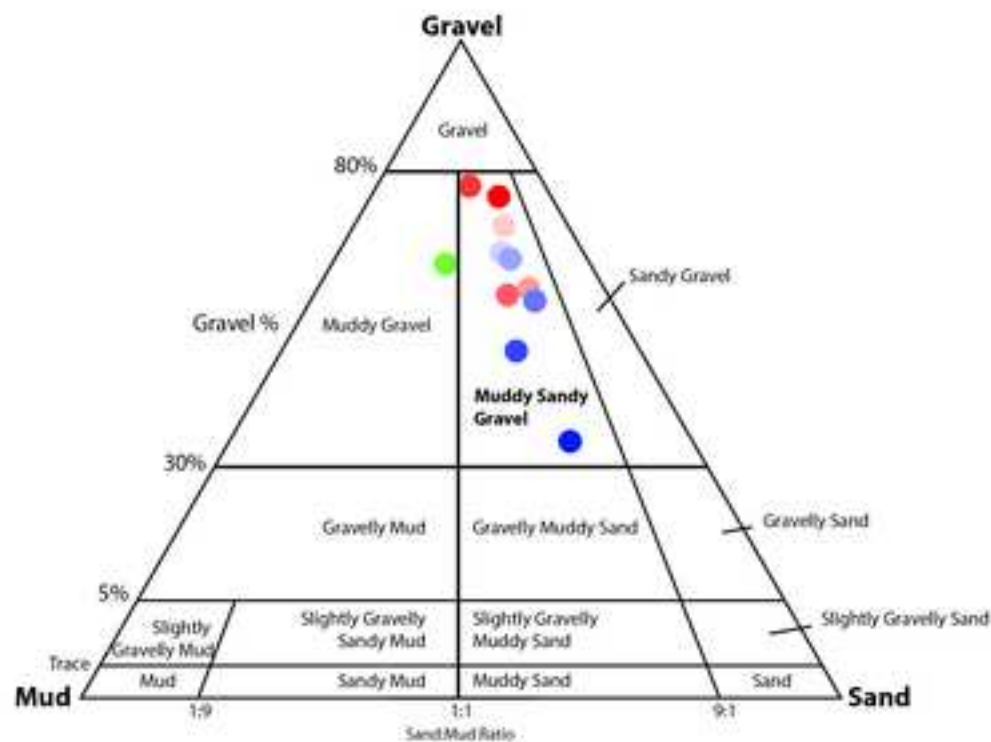
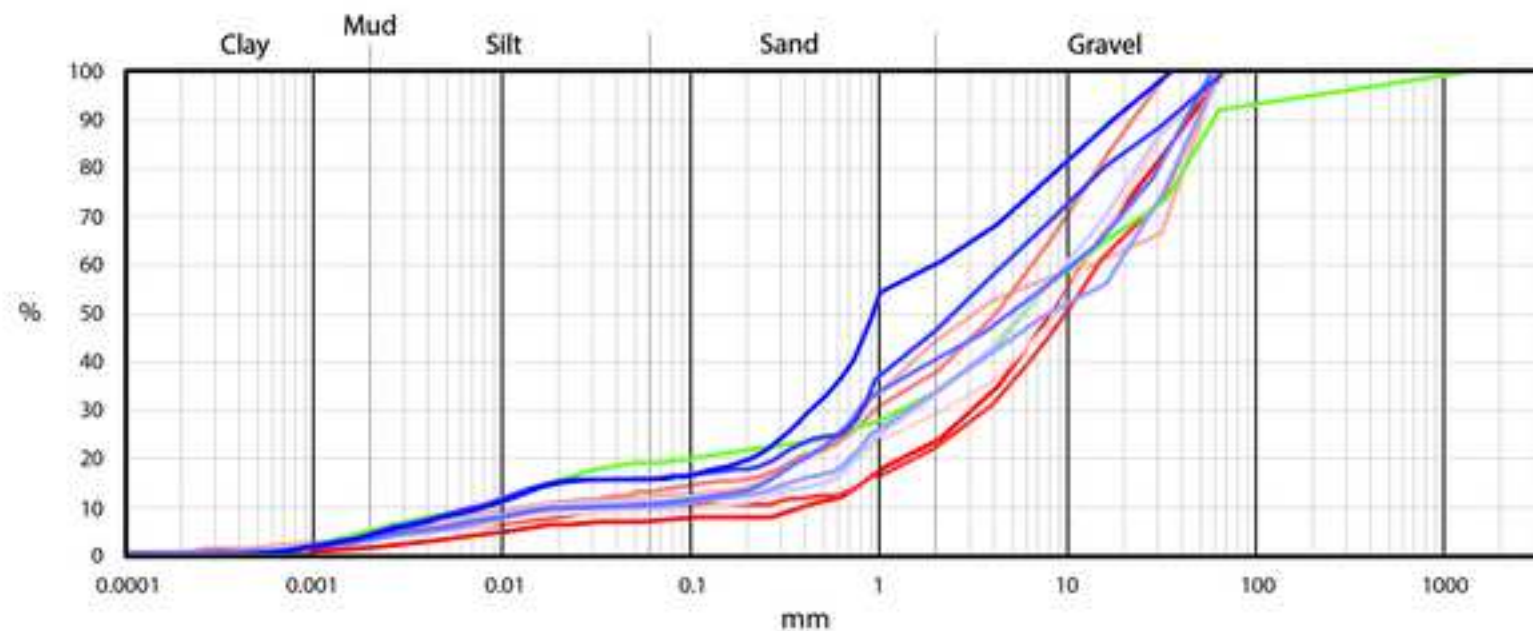


Figure 10

